

Jiahao Zheng

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EDUCATION

Nanjing University of Posts and Telecommunications

Sep 2023 – Present

B.E. in Internet of Things Engineering, School of Internet of Things Engineering

GPA: 4.12/5.00 Rank: 1/103

- **Selected Coursework:** Digital Circuits and Logic Design (100), Data Structures and Algorithms (98), College Physics (97), Advanced Mathematics (96), Discrete Mathematics (96), Programming (95), Computer Networks (95). More than **25 courses scored above 95**.
- **English:** CET-4, CET-6 (**498**).
- **Research Interests:** 3D Vision & Spatial Intelligence, Multimodal LLMs, World Models, Embodied AI.
- **Honors:** **National Scholarship** (sole recipient in major) · National Encouragement Scholarship · **Cilin Scholarship** · Outstanding Student Model (top 1%) · Outstanding Communist Youth League Member.

REPRESENTATIVE PUBLICATIONS

Technical trajectory: from sparse-view 3D reconstruction to multimodal editable generation and large-scale Real2Sim scene instantiation, then to robot perception and Sim2Real deployment.

- [1] Jianhui Zheng, **Jiahao Zheng**, Zhiyong Huo*. “Towards Robust Sparse-View 3D Gaussian Splatting via Hierarchical Depth and Multi-View Consistency.”

ACM International Conference on Multimedia Retrieval, 2026. **CCF-B, 2nd Author**.

Project: National Undergraduate Innovation Project on robust sparse-view 3D Gaussian Splatting, targeting geometric degradation, texture overfitting, and structural drift through hierarchical depth priors, foreground semantics, random Gaussian rendering, and multi-view consistency.

My role: Contributed semantic-driven hierarchical depth supervision and random Gaussian rendering; responsible for part of training, baseline reproduction, and evaluation. Average **PSNR reached 12.60/18.22/22.44 dB** and **LPIPS reached 0.490/0.211/0.135**, respectively. One invention patent granted.

- [2] **Jiahao Zheng**, Shu Liu, et al. “Diamond Thermometry Beyond the Zero-Phonon Line via Physics-Aware Spectral Trend Learning.”

Applied Physics Letters, 2026. **JCR Q1, Nature Index, 1st Author**.

Project: Physics-aware NV thermometry under ZPL quenching; models phonon-sideband evolution, spectral-trend continuity, and cross-sample calibration constraints for robust temperature inference when the zero-phonon line becomes unreliable.

My role: Independently designed the physics-constrained phonon-sideband ResNet thermometry network, built the spectral preprocessing and ablation pipeline, completed transfer experiments, and achieved **0.045 K** after cross-sample fine-tuning.

- [3] Honghao Chen, Haoyu Li, **Jiahao Zheng**, et al. “Real-Time On-Chip Nanosecond Pulse Capture with NV Centers: U-Net Denoising and Signal Reconstruction.”

Chinese Physics B, 2026. **JCR Q3, 3rd Author**.

Project: Real-time NV-center nanosecond pulse readout and signal reconstruction under extremely low SNR, combining U-Net denoising, digital signal inversion, pulse-pair calibration, and physics-aware constraints for on-chip waveform recovery.

My role: Designed U-Net physics-aware loss and digital signal inversion, built evaluation on **40,000** pulse pairs, and optimized the denoising–reconstruction pipeline, improving SNR by **21.39 to 29.57 dB** and reducing BER to **6.12%**. Related team work was nominated for the **2025 IoT Conference Top 10 Advances**.

RESEARCH EXPERIENCE

Editable CAD Generation via Multimodal LLMs

Mar 2026 – Present

Tsinghua University GME Lab | NeST, AAAI 2027 submission, 2nd Author

- Participated in **NeST**, a multimodal editable CAD generation framework that maps text, views, and point clouds into executable modeling programs with **Qwen3-VL-4B** and **Cadrille**.
- Introduced high-dimensional structural representations that encode geometry, topology, and parametric relations, improving over raw XYZ point-cloud descriptions for complex CAD constraints.
- Designed CAD-kernel-feedback rewards and constraint-aware generation alignment; built a **12-dim constrained CAD dataset**, with point-cloud input reaching **IoU > 92%** on DeepCAD.

Real2Sim: Interactive Urban Scene Instantiation

Nov 2025 – Present

Institute of Computing Technology, Chinese Academy of Sciences, IIP-VIPL Lab

Beijing, China

Supervisors: Prof. Shiguang Shan & Zhenliang He

- Built a multimodal semantic-driven retrieval-generation framework for **UrbanVerse**, addressing occlusion, long-tail asset missingness, interaction deficiency, and pose drift in Real2Sim urban reconstruction.
- Recovered instance-level layouts through multi-frame semantic-geometric consistency, introduced semantic direction constraints for 6D pose estimation, and triggered generative asset completion under low retrieval confidence.
- Integrated semantic and physical priors with layout-rule constraints to convert road videos into editable **UE5** interactive scenes; largely completed Zhongguancun public-road reconstruction.

Robotic Arm Control via MuJoCo Simulation and Sim2Real

Jul 2025 – Aug 2025

Tsinghua University AIR Summer Program

- Participated in a MuJoCo-based Sim2Real manipulation pipeline for AIRBOT Play, covering simulator construction, ACT policy training, real-robot deployment, and generalization optimization.
- Connected perception, task understanding, organizing, and grasping into a closed-loop food-model manipulation workflow, with robustness improved through simulation-to-real transfer strategies.
- Reached **over 90%** real-robot grasp success and **50%** single-trial dual-target grasping; awarded **Best Team** and **Outstanding Camper**.

ENGINEERING PROJECTS & OPEN SOURCE

Part2FullNet: Sentry Robot Vision for RoboMaster RMUL 2026

Oct 2024 – Mar 2026

NJUPT Potential Team, Vision Group | 3rd Place, Jiangsu Regional

- Built a sentry-robot vision module for robust armor-plate perception under severe occlusion, where missing corners make conventional PnP unstable.
- Proposed **Part2FullNet**, a lightweight residual completion network fusing monocular depth, rigid geometric constraints, and **EKF temporal priors**; used a 3-layer MLP to recover full-corner residuals.
- Coupled completed corners with **IPPE-PnP** for 6D pose solving, achieving **over 75%** hit rate on highly maneuvering occluded targets and **below 7.5 mm** lateral PnP localization error.

Clutter Terminator: RISC-V Intelligent Desktop Organizer

Nov 2024 – Dec 2025

National College Student Integrated Circuit Innovation & Entrepreneurship Competition

- Customized YOLOv5s on RISC-V EIC7700 NPU for real-time object detection in cluttered-desktop scenarios, achieving **mAP = 97.4%**.
- Built a DeepSeek-R1-based intelligent agent that perceives desktop objects, reasons about spatial layout, and plans optimal tidying sequences via a remote interaction system.
- Integrated HAE and DSP-accelerated operators to form a real-time closed loop with inference latency reduced to **3 ms**. Software copyright registered.

Open-Source Projects github.com/jiahaozheng406

- **3DGS-Research** – Codebase and experiments for sparse-view 3D Gaussian Splatting research (SegGeoGS).
- **Cityscapes-Seg** – Urban semantic segmentation with Mask2Former on Cityscapes.
- **Quadrotor-Trajectory-Tracking** – PD + feedforward controller for quadrotor trajectory tracking in Simulink.
- **Voice-LLM-SCARA-Robot** – Voice-controlled SCARA robotic arm with LLM-based intelligent interaction on RISC-V.
- **Physics-Agent** – LLM-powered physics problem-solving agent with tool-use and reasoning capabilities.
- **Basic-Image-Filtering** – Classical Gaussian, median, and bilateral filters from scratch.

COMPETITIONS & AWARDS

Most awards below obtained as **Team Leader**.

Second Prize (National) , National IC Innovation & Entrepreneurship Competition	2025
3rd Place (National) , RoboMaster 2026 RMUL, Jiangsu Regional	2026
Third Prize (National) , 2026 China Robot Competition & RoboCup China Open	2025
Meritorious Winner , Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling	2025
Second Prize (National) , National Undergraduate Mathematics Competition	2025
Third Prize (National) , iCAN College Student Innovation & Entrepreneurship Competition	2024
Third Prize , Asia and Pacific Mathematical Contest in Modeling (APMCM)	2025
First Prize (Provincial) , Global Campus AI Algorithm Elite Competition (AIC)	2025
Second Prize (Provincial) , National Physics & Experimental Technology Innovation Competition	2024

LEADERSHIP & SOCIAL PRACTICE

President & Co-Founder, Innovation and Entrepreneurship Association, NJUPT Sep 2024 – Present

- Co-founded the association; attracted 10+ angel investors/VC partners, organized 50+ university–industry events, connecting 300+ alumni.
- Planned and delivered 20+ campus events on frontier technologies and research competitions, serving 10,000+ students cumulatively.

Social Practice & Volunteering

- Represented NJUPT at the 21st Jiangsu Provincial Games in 10 000 m run and 10 000 m race walking events.
- 2024 National Youth Science Camp – Outstanding Volunteer, organized and guided science activities for young students.
- Nanjing University Student Track and Field Championship – represented NJUPT in competitive athletics.
- Summer Social Practice in Yan’an – Outstanding Team, led field research and community engagement in revolutionary heritage sites.

SKILLS & ADDITIONAL INFORMATION

- **Tools & Platforms:** Linux, PyTorch, Python, HTML, Git, LaTeX, Origin, Visio, MATLAB, MySQL, PowerPoint.
- **Additional:**
 - Strong research-driven mindset with solid theory and system implementation foundations.
 - Independent research ability, rapid learning, and effective cross-disciplinary collaboration.
 - Excellent scientific illustration and research presentation skills.
- **Hobbies:** Marathon, Football, Climbing.